Lab 10 – External Flow over a Cylinder

Grant Bellon

Environmental Conditions

Quantity	Value	Uncertainty
Barometric pressure, P_{atm}	999.7 mbar	± 0.1 mbar
Room temperature, T_{atm}	27.1 °C	± 1 °C

Water Density Sample

Quantity	Value	Uncertainty
Water volume, Vol_{water}	24 mL	± 0.5 mL
Water mass, m_{water}	24.2 g	± 0.05 g

Geometry

Quantity	Value	Uncertainty
Cylinder diameter, D	24.9 mm	± 0.1 mm
Cylinder length, $L_{cylinder}$	5 7/8 in	± 1/32 in
Water channel width, $w_{channel}$	6 1/4 in	± 1/32 in
Wetted height, h_{wetted}	5 1/8 in	± 1/32 in
L_1	2 1/16 in	
L_2	6 1/4 in	± 1/32 in

Drag Force Runs

Motor controller setpoint	Flowrate, Q	U_Q	Tension, T	U_T
10	52.7 G/min	± 0.5 G/min	0.146 N	± 0.01 N
20	106.2 G/min	± 0.5 G/min	1.02 N	± 0.01 N
30	162.8 G/min	± 2 G/min	2.86 N	± 0.1 N
40	208.2 G/min	± 3 G/min	3.68 N	± 0.1 N
50	223.5 G/min	± 5 G/min	3.98 N	± 0.1 N
60	247.8 G/min	± 7 G/min	4.73 N	± 0.2 N